

and thus the typical participant in the equity market has a modest degree of risk aversion.

The mean-variance solution in equation (2.10) turns out to be the same as CRRA utility (see equation (2.1)) if returns are log-normally distributed. This is one sense that mean-variance and CRRA are equivalent.

4. Realistic Utility Functions

There are several shortcomings of mean-variance utility:

1. The variance treats the upside and downside the same (relative to the mean).
However, asset owners generally exhibit asymmetric risk aversion, feeling losses more powerfully than any equivalent gain.
2. Only the first two moments matter.
People prefer positive skewness, like the chance of a big lottery payoff, and dislike negative skewness, like the barreling steamroller crushing you as you pick up nickels and dimes.
3. Subjective probabilities matter.
The way that people perceive probabilities differs from the actual distribution of returns. In particular, people tend to overestimate the probability of disasters.
4. Bad times other than low means and high variances matter.
Bad times aren't just times of low wealth. Mean-variance utility rather simplistically portrays bad times as consisting only of low means and high variances. But being rich or poor in absolute terms might not matter so much as whether you are rich or poor relative to your neighbor or whether you are rich or poor now relative to whether you were rich or poor in the past. That is, your utility could be relative.

There are many richer models of utility that incorporate all these considerations. Some of these models fit into the expected utility framework of the previous sections, so all of the previous intuition and economic machinery apply. In economics, there are many utility functions that realistically describe how people behave. In the asset management industry, unfortunately, only one utility model dominates—by a long shot, it's the restrictive mean-variance utility model. It would be nice if we had a commercial optimizer where one could toggle between various utility functions, especially those incorporating downside risk aversion. It would be even better if an application could map a series of bad times, and how the risk of these bad times is perceived by an investor, to different classes of utility functions. Sadly, there are no such asset allocation applications that I know of at the time of writing that can do this. And yet all the economic theory and optimization techniques are already published.